

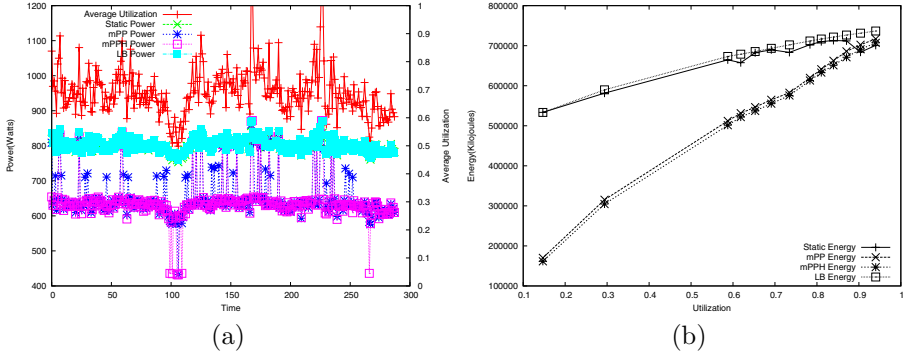


Fig. 7. Power consumed by various placements strategies with (a) time and (b) increasing overall utilization

observe that the dynamic algorithms *mPP* and *mPPH* are able to save about 200W of power (25%) from the Load Balanced and Static Placements. However, as the utilization exceeds 0.75, the savings drop significantly. This is because, at high utilization, there is not much scope for consolidation. However, even at high loads, our algorithms are able to save some power by taking the decreasing slope of the power curves into account. We observed that our algorithms try to run most servers at close to their full capacity because the power-curve saturates at high capacity. Hence, instead of balancing the load across all servers, it makes sense to unbalance load even when all the servers are switched on. We also observe that the proposed algorithms show very strong correlation with the average utilization. This establishes their ability to continually adapt to workload variations and save power accordingly.

We next investigate the impact on power at different average utilization values. Towards this purpose, we compressed (or decompressed) all the traces so that the average utilization of the servers could be varied. The trace was compressed to achieve a higher utilization than the baseline, and decompressed to achieve lower utilizations. We observed that the power savings obtained by the proposed power-minimizing algorithms is significant at low utilization values. This is because our algorithms are able to consolidate the VMs on a few servers only, and save a significant amount of power due to the high static power cost. Our study (Fig. 7(a)) reaffirms that most savings come from consolidation and not by unbalancing load. This is evident because the savings of proposed algorithms increase significantly when the average utilization goes below 0.75 and 0.5. These are the utilization values at which one can switch off an additional server. Note that the discontinuities are not so noticeable in Fig. 7(b). However, since at an average utilization of 0.75, there are many time instances where the utilization goes beyond 0.75 and no server can be switched off, the difference between overall energy consumption between average utilization of 0.7 and 0.75 is not as large as one would expect from Fig. 7(a).

We have only investigated the performance of algorithms w.r.t power consumed by their corresponding placements. We now take into account the